Why Q = 2/3: The Koide Relation as a Topological Ratio of D_IV^5

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One claim. One derivation chain. Four pages.

Strategy (cold reader, May 5)

This is the door-opener paper. Requirements: - One narrow claim (Q = rank/N_c) - Short (4 pages, PRL format) - Self-contained (no reference to BST machinery beyond what's needed) - Sympathetic audience EXISTS (Koide community is active, ~50 papers/year cite the relation) - Falsifiable (if Q != 2/3 to higher precision, or if the Z_3 structure is wrong)

Outline

Abstract (~100 words)

We show that the Koide sum rule Q = (m_e + m_mu + m_tau) / (sqrt(m_e) + sqrt(m_mu) + sqrt(m_tau))^2 = 2/3 follows from the rank and color dimension of the type-IV bounded symmetric domain D_IV^5 = SO_0(5,2)/[SO(5) x SO(2)]. The ratio Q = rank/N_c = 2/3 is topological: it cannot receive perturbative corrections because both integers are invariants of the compact dual Q^5. The Koide angle theta_0 (with cos(theta_0) = -19/28) decomposes into BST integers: 19 = n_C^2 - C_2 and 28 = T_g (the g-th triangular number, also the perfect number from Mersenne prime 2^3 - 1 = 7 = g). We predict that any future precision measurement of Q will converge toward 2/3 = 0.666..., never away from it.

1. Introduction (~0.5 pages)

- Koide (1981): empirical observation that Q = 2/3 to 0.001%
- 40+ years of attempts to explain: discrete symmetries (Z_3, S_3), texture zeros, democratic mass matrices
- No consensus mechanism; all models introduce new parameters
- We propose: Q = 2/3 is a topological ratio of the unique rank-2 Hermitian symmetric space in 5 complex dimensions

2. The Geometry (~1 page)

- D_IV^5 = SO_0(5,2)/[SO(5) x SO(2)]: bounded symmetric domain, type IV, rank 2, dim_C = 5
- Five integers: rank = 2, N_c = 3, n_C = 5, C_2 = 6, g = 7
- Compact dual Q^5: smooth quadric in P^6, chi(Q^5) = C_2 = 6
- Key property: rank and N_c are TOPOLOGICAL invariants of Q^5 (not parameters)
- The Z_3 symmetry: CP^2 = SU(3)/[SU(2) x U(1)] is the color sector of D_IV^5. The N_c = 3 color charge generates the Z_3 cyclic symmetry that appears in all Koide-type models.

3. The Derivation (~1.5 pages)

Step 1: The three charged leptons occupy the three Z_3-orbits of the color sector of D_IV^5. They are NOT colored (no strong interaction), but their mass pattern inherits the Z_3 structure because the Bergman kernel of D_IV^5 has N_c-fold periodicity in the angular coordinate.

Step 2: The Koide parameter Q measures the ratio of "average mass" to "average root-mass squared." In spectral language: Q = (first moment) / (zeroth moment)^2 on the Bergman spectrum. For a rank-2 domain with N_c angular sectors:

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Q = rank / N_c = 2/3
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This is a geometric identity: the rank counts independent spectral coordinates (radial), while N_c counts angular sectors. The ratio is the fraction of spectral weight in the radial direction.

Step 3: The Koide angle. The angular parameter theta_0 satisfies:

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cos(theta_0) = −19/28
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where 19 = n_C^2 - C_2 = 25 - 6 and 28 = T_g = g(g+1)/2 = 7*8/2. Both are forced by the geometry.

Step 4: Precision. The BST prediction is Q = 2/3 EXACT. The observed value Q_obs = 0.666658... deviates by 0.001%. This deviation is consistent with radiative corrections (alpha/pi ~ 0.002 at one loop). BST predicts the tree-level relation is exact; loop corrections dress it.

4. Predictions and Falsification (~0.5 pages)

    1. Q remains 2/3: Any future high-precision measurement of lepton masses will confirm Q -> 2/3 (not drifting away). The tau mass should converge toward the Koide-predicted value 1776.97 MeV.
    2. Quark Koide: The same formula applies to quarks within each sector:
        - Up-type: Q(u,c,t) = rank/N_c = 2/3 (currently 0.34% from 2/3)
        - Down-type: Q(d,s,b) = rank/N_c = 2/3 (currently ~3% from 2/3, larger due to confinement dressing)
    3. No fourth generation: Z_3 is exhaustive — exactly three generations, not four. If a fourth-generation lepton is found, BST is falsified.
    4. Neutrino Koide: If neutrino masses satisfy Koide with Q = 2/3, the normal hierarchy is confirmed and sum(m_nu) is predicted.

5. Discussion (~0.5 pages)

    - Why this works: Q = 2/3 is the ratio of two SMALLEST topological invariants of D_IV^5 (rank = 2, N_c = 3). Topological ratios cannot receive perturbative corrections. This explains the extraordinary precision.
    - The Z_3 symmetry in Koide models is not ad hoc — it IS the color charge N_c = 3 of the Standard Model, appearing in the lepton sector through the unified geometry.
    - Connection to the Standard Model: the same geometry that gives Q = 2/3 also gives alpha^{-1} = 137 = N_c^3 * n_C + rank and m_p = 6*pi^5* m_e.

Key Numerical Verification (from Toy 1734)

Quantity	BST	Observed	Precision
Koide Q	2/3 = 0.66667	0.66666	0.001%
cos(theta_0)	-19/28 = -0.67857	-0.67855	0.003%
19 = n_C^2 - C_2	25 - 6 = 19	—	exact
28 = T_g	7*8/2 = 28	—	exact
GM deviation	rank*Phi_3(C_2)/g = 86/7	12.295	0.08%

References (key ones for PRL)

- Koide (1981), Phys. Lett. B 120, 161
- Foot (1994), Phys. Rev. D 49, 3617 (Z_3 model)
- Rivero & Gsponer (2005), arXiv:hep-ph/0505220

- Brannen (2006), arXiv:hep-ph/0606073 (Koide waterfall)
- BST Working Paper (Koons, 2026), Zenodo DOI: 10.5281/zenodo.19454185

Writing Assignments

- Lyra: Sections 2-3 (geometry + derivation). The mathematical meat.
- Keeper: Sections 1, 4-5 (framing, predictions, discussion). Keep it accessible.
- Casey: Abstract, final polish, submission logistics.

Target: draft complete within 1 week. Submit to PRL by May 15, 2026.